Daniel Hardesty Lewis

Staff ML Engineer – production training, inference, evaluation, reliability

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

Core Expertise

8 years	**ML Engineering**: Build and operate AI/ML training and inference pipelines, with evaluation harnesses that catch regressions and improve production reliability
8 years	**Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS
8 years	**Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency/throughput with explicit cost/perf tradeoffs
10 years	**Geospatial AI**: Build spatial-temporal (geospatial + time-series) pipelines and models for risk, planning, and hazard decision-support use cases
8 years	**Technical Leadership**: Drive execution through specs and stakeholder alignment; mentor engineers through reviews and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Hugging Face Transformers; LoRA/QLoRA/PEFT
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase
Infra	Linux/Unix; Bash; Git; Docker; Kubernetes; Ray; Spark; Dask; Airflow; CI/CD; observability
Additional	CUDA; C++; TypeScript; React

Experience

Professional

Apr. 2025 – Present **Co-founder**, *Properlytic*, New York, NY

Building a consumer real-estate pricing product focused on individual home price forecasting.

- Deployed a current-price prediction model for Manhattan condos and co-ops; achieved 12% MAPE on a holdout set and matched Zillow at 8.4% MAPE under an internal apples-to-apples evaluation protocol.
- Architected a VAE-based latent representation layer over property, transaction, and neighborhood signals to stabilize learning and improve generalization under sparse and noisy features.
- Prototyped a diffusion-based forecasting backbone in Houston; measured 8% annualized compounding error even through regime shifts such as COVID.
- Migrated market-specific models onto the diffusion backbone for standardized training and evaluation.
- Refined product positioning and execution milestones via strategic feedback from a16z, MetaProp, and others.
- Pitch deck: link

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow and schema definitions for BOM parsing, retrieval, and tariff simulation using Airflow, Elasticsearch and embeddings, and FastAPI serving backed by Supabase and Redis.
- Engineered a what-if framework to analyze BOM cost impacts, lead times, and compliance risks.
- Secured market validation by presenting the prototype to KPMG, pre-dating their subsequent launch of a commercial tariff-modeling product.
- Executing customer discovery with global importers/exporters to iterate on distribution and compliance pricing models.

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows.

- Engineered a 0.5 m statewide seamless Texas DEM mosaic using nearest-neighbor stitching across 70+ state, federal, and local elevation sources.
- Developed the web distribution platform end-to-end (data pipeline, tiling, hosting, delivery UX) to support engineering and planning use cases.
- Commercial discussions to license elevation datasets to research and AI organizations.

155 Claremont Ave – New York, NY 10027 – United States

+1 (713) 371-7875 • *daniel@summitgeospatial.com* • *dlewis.ai* • *dhardestylewis*

dhardestylewis • *Google Scholar*

Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Co-founded and launched summitgeospatial.com to commercialize state-level geomorphology IP, providing precision decision-support for civil engineering and disaster planning.
- Led the source-to-frontend data pipeline for the $40M Texas Disaster Information System (TDIS), integrating large-scale geospatial assets for state-level resiliency.
- Scaled climate and flood models on supercomputers, executing million-node distributed jobs while managing multi-million-dollar compute budgets and federal partnerships (NOAA, NSF, USCE, GLO).
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Developed efficient methods to produce high-resolution (1m) flood maps from National Water Model outputs for Texas Emergency Management (Real-Time Inundation Mapping).
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.

Research

May 2025 – Present **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

- Targetting theory publications to DeepSeek's GRPO algorithm

Jan. 2025 – Present **Research assistant to Dir. Financial Engineering Ali Hirsa**, *Columbia University*, New York, NY

- Developed a multi-asset conditional variational autoencoder (CVAE) latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

- Helped teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, HPC, & optimization.

Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

Activities

Transatlantic Sailing Volunteer offshore SAR (Cheeki Rafiki)

Bikepacking Across U.S. East and West coasts

Languages

Spanish Native Proficiency

French Professional Working Proficiency

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